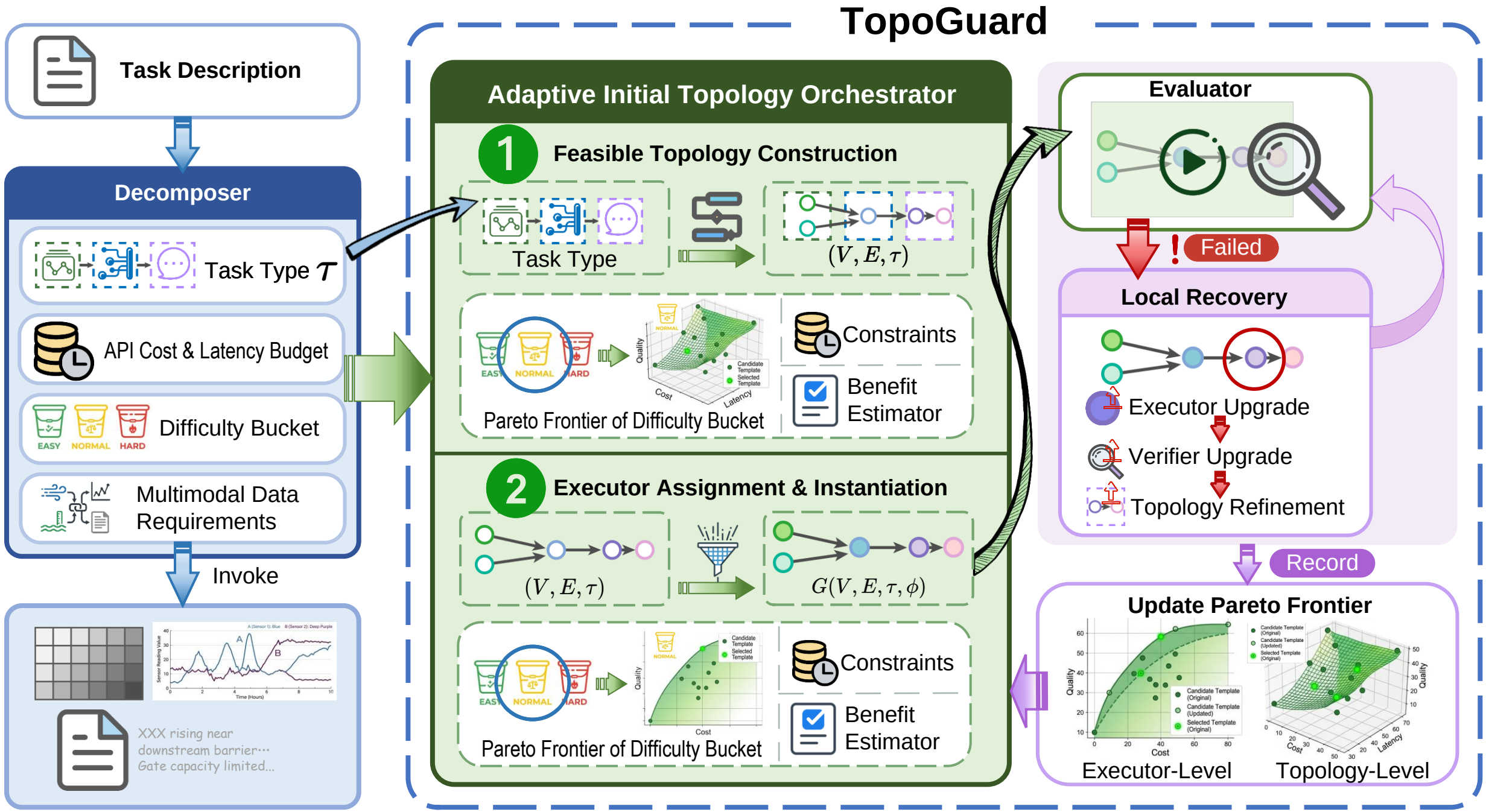
TopoGuard
Task Description
Decomposer
Task Type $\mathcal{T}$
API Cost & Latency Budget
Difficulty Bucket
EASY
NORMAL
HARD
Multimodal Data Requirements
Invoke
XXX rising near downstream barrier…
Gate capacity limited…
Adaptive Initial Topology Orchestrator
1 Feasible Topology Construction
Task Type
(V, E, τ)
Pareto Frontier of Difficulty Bucket
Constraints
Benefit Estimator
2 Executor Assignment & Instantiation
(V, E, τ)
$G(V, E, \tau, \phi)$
Pareto Frontier of Difficulty Bucket
Constraints
Benefit Estimator
Evaluator
Failed
Local Recovery
Executor Upgrade
Verifier Upgrade
Topology Refinement
Record
Update Pareto Frontier
Executor-Level
Topology-Level